

Monitoring and Management

What happened, who did it, and what the configuration was. Three different services.

How is it behaving? Who did it? What was the configuration? Three questions, three services.

The three-way separator

SERVICE	THE QUESTION IT ANSWERS	TYPICAL EVIDENCE
Amazon CloudWatch	How is it performing?	CPU at 90%, a failed status check
AWS CloudTrail	Who did that, and when?	User Dan called DeleteBucket at 14:02
AWS Config	What was it set to?	This bucket was public between March and June

ON THE EXAM

A metric over time is CloudWatch. An identity performing an action is CloudTrail. A resource's settings are Config.

Amazon CloudWatch

- ◆ **Metrics** are numeric measurements over time. Most AWS services publish them automatically.
- ◆ **Logs** collects application and system log files you send to it, and can query them.
- ◆ **Alarms** watch a metric and act when it crosses a threshold you set.
- ◆ Dashboards combine metrics from several services, Regions and accounts into one view.
- ◆ EC2 does not report memory or disk usage by default. Those need the CloudWatch agent installed.

ON THE EXAM

The memory metric is a favourite exam detail. Memory usage requires the CloudWatch agent; CPU does not.

CloudWatch alarms in practice

- ◆ **An alarm has three states:** OK, ALARM, and INSUFFICIENT_DATA.
- ◆ Alarm actions can notify through Amazon SNS, trigger Auto Scaling, or stop and reboot an instance.
- ◆ **Alarms are how monitoring becomes automation.** Detection alone changes nothing.
- ◆ Billing alarms watch estimated charges, though AWS Budgets is the better answer for cost questions.
- ◆ Amazon EventBridge routes events, where CloudWatch alarms react to thresholds. Both can start a response.

ON THE EXAM

Notify someone when a metric crosses a value means a CloudWatch alarm plus SNS. That pairing appears constantly.

AWS CloudTrail

- ◆ **Records API calls across AWS**, whoever made them and however they were made.
- ◆ Console clicks, CLI commands, SDK calls and CloudFormation all become API calls, so all are recorded.
- ◆ **The last ninety days of management events are viewable free** in the console as Event history.
- ◆ Creating a trail delivers events to Amazon S3 for indefinite retention, and can cover all Regions.
- ◆ Log file integrity validation proves the log has not been altered, which auditors ask for.

ON THE EXAM

Which user made this change is always CloudTrail. If an identity is named in the question, that is the signal.

CloudTrail or CloudWatch?

AWS CLOUDTRAIL

Who did what

- Records API calls and the identity behind them
- Answers which user changed a security group
- Auditing, governance and compliance
- The subject is an action by someone

AMAZON CLOUDWATCH

How it is behaving

- Collects metrics, logs and alarms
- Answers whether an instance failed a status check
- Monitoring and operational health
- The subject is a measurement over time

THE TIEBREAKER

If the question names a user or asks who, it is CloudTrail. If it asks how much, how fast or whether something failed, it is CloudWatch.

AWS Config or CloudTrail?

AWS CONFIG

What it looked like

- Records resource configuration over time
- Answers was this bucket ever public
- Evaluates rules for compliance
- The subject is a state

AWS CLOUDTRAIL

Who changed it

- Records the API call and the caller
- Answers who made the bucket public
- Auditing of actions
- The subject is an event

THE TIEBREAKER

Config answers what the configuration was over time. CloudTrail answers who changed it. Compliance rules always mean Config.

AWS Trusted Advisor

- ◆ **Continuously checks your account against AWS best practices** and recommends changes.
- ◆ **Six categories:** cost optimisation, performance, security, fault tolerance, service limits, and operational excellence.
- ◆ Older material says five. Operational excellence was added in October 2023; the other five are unchanged.
- ◆ Every account gets service limit checks and a small set of security and fault tolerance checks free.
- ◆ The full check set requires a paid support plan. Trusted Advisor Priority requires Enterprise.

ON THE EXAM

If a question offers five categories and they are the original five, that answer is still correct. Do not overthink it.

Trusted Advisor or Compute Optimizer?

AWS TRUSTED ADVISOR

Broad best-practice checks

- Six categories across the whole account
- Idle resources, open ports, approaching quotas
- AWS best practices, not your rules
- Breadth over depth

AWS COMPUTE OPTIMIZER

Rightsizing, using measurements

- EC2 instances, Auto Scaling groups, EBS volumes, Lambda
- Analyses actual utilisation metrics
- Recommends a specific different size
- Depth on one question only

THE TIEBREAKER

Is this resource the right size for its usage means Compute Optimizer. Anything broader means Trusted Advisor.

AWS Systems Manager

- ◆ **Patch Manager** applies operating system patches across a fleet on a schedule.
- ◆ **Session Manager** gives a shell on an instance with no SSH keys, no bastion host and no open inbound ports.
- ◆ **Parameter Store** holds configuration values and secrets, encrypted with KMS.
- ◆ Run Command executes a command across many instances at once without logging in.
- ◆ It works on EC2 instances and on servers outside AWS, which makes it a hybrid management tool.

ON THE EXAM

Access an instance without opening port 22 or managing keys is the Session Manager scenario, and it appears often.

AWS Health Dashboard

- ◆ **Service health** shows the general status of AWS services, publicly, for everyone.
- ◆ **Your account health** shows events affecting your specific resources and accounts.
- ◆ It notifies you of scheduled maintenance, retirements and issues touching resources you actually own.
- ◆ The AWS Health API allows programmatic access and integration with your own tooling.
- ◆ It reports on AWS's health, not on your application's. Your application is CloudWatch's job.

ON THE EXAM

Is AWS having a problem, or scheduled maintenance affecting my instances, means the Health Dashboard.

Three more worth recognising

- ◆ **AWS X-Ray** traces a request through a distributed application to find where the time is going.
- ◆ The signal is microservices, and not knowing which component is slow.
- ◆ **Service Quotas** shows your quotas and lets you request increases from one place.
- ◆ **AWS License Manager** tracks software licence usage against your agreements.
- ◆ Each has one distinctive signal. None of them appears often, but each is unmistakable when it does.

ON THE EXAM

Find the bottleneck in a microservices application is X-Ray. Nothing else in AWS does distributed tracing.

Reverse lookup

IF THE SCENARIO SAYS	THE ANSWER IS
Which user deleted the resource	AWS CloudTrail
Alert me when CPU exceeds a threshold	Amazon CloudWatch alarm with SNS
Was this resource ever configured incorrectly	AWS Config
Recommendations to cut cost and improve security	AWS Trusted Advisor
Is this instance the right size for its load	AWS Compute Optimizer
Connect to an instance without opening SSH	Systems Manager Session Manager
Is AWS having an incident affecting us	AWS Health Dashboard
Which microservice is causing the delay	AWS X-Ray

ON THE EXAM

Eight phrasings that cover nearly every question in this area. The first three carry the most marks.

An Amazon S3 bucket holding customer records was found to be publicly readable. The security team needs to establish three things: which IAM identity changed the bucket policy, whether the bucket had ever been public before this incident, and whether anyone was alerted at the time.

The company runs CloudTrail, Config and CloudWatch.

Which service answers the first two questions, in order?

- | | |
|--|--|
| ☐ A CloudWatch, then CloudTrail | ☐ B CloudTrail, then AWS Config |
| ☐ C AWS Config, then CloudWatch | ☐ D CloudTrail, then CloudWatch |

◆ PAUSE AND DECIDE

Which service answers the first two questions, in order?

- A

 CloudWatch, then CloudTrail
CloudWatch holds metrics and logs. It does not record who called an API, nor a resource's past configuration.
- B

 CloudTrail, then AWS Config
Which identity made the call is CloudTrail. Whether it was ever public before is configuration history, which is Config.
- C

 AWS Config, then CloudWatch
Both halves wrong. Config does not name the caller, and CloudWatch holds no configuration history.
- D

 CloudTrail, then CloudWatch
First half right, second wrong. CloudWatch cannot tell you what a bucket's policy was three months ago.

THE SIGNAL

Two questions, two services. Which identity is always CloudTrail; ever been in this state is always Config.

An operations team wants to be notified when any EC2 instance in production sustains CPU utilisation above eighty percent for fifteen minutes, so an engineer can investigate before users are affected.

They also want the notification to reach an on-call email group.

Which combination meets this requirement?

- ☐ A AWS CloudTrail with an S3 trail
- ☐ B A CloudWatch alarm with an Amazon SNS topic
- ☐ C AWS Config with a custom rule
- ☐ D AWS Trusted Advisor with weekly refresh

◆ PAUSE AND DECIDE

Which combination meets this requirement?

- A

 AWS CloudTrail with an S3 trail
Records API calls. CPU utilisation is not an API call and CloudTrail never sees it.
- B

 A CloudWatch alarm with an Amazon SNS topic
The alarm watches the metric and the threshold; SNS delivers the email. The standard pairing.
- C

 AWS Config with a custom rule
Evaluates configuration, not live performance. CPU utilisation is not a configuration property.
- D

 AWS Trusted Advisor with weekly refresh
Periodic best-practice checks. It cannot alert on a threshold within fifteen minutes.

THE SIGNAL

A numeric threshold sustained over a period is a metric, and metrics are always CloudWatch.

A company must patch operating systems on two hundred EC2 instances every month. It also wants engineers to be able to open a shell on any instance for troubleshooting, but security policy forbids opening inbound SSH ports and forbids distributing SSH keys. The company wants one service to cover both needs.

Which service should the company use?

- ☐ A AWS Systems Manager
- ☐ B Amazon CloudWatch
- ☐ C AWS Config
- ☐ D A bastion host in a public subnet

◆ PAUSE AND DECIDE

Which service should the company use?

- A** AWS Systems Manager
Patch Manager handles the patching; Session Manager gives shell access with no open ports and no keys.
- B** Amazon CloudWatch
Monitors and alerts. It neither patches instances nor provides access to them.
- C** AWS Config
Could report which instances are unpatched, but reporting is not patching.
- D** A bastion host in a public subnet
Requires exactly what the policy forbids: an open inbound port and SSH keys to distribute.

THE SIGNAL

Two requirements, one service. Session Manager exists precisely to remove bastion hosts and SSH key management.

A company's application suddenly slows down. Engineers confirm their own instances are healthy, with normal CPU and no errors in application logs. They suspect an issue with an AWS service in their Region.

They want to know whether AWS is experiencing a problem that affects their specific resources, and whether any maintenance is scheduled.

Where should they look?

- ☐ A Amazon CloudWatch dashboards
- ☐ B The AWS Health Dashboard
- ☐ C AWS CloudTrail event history
- ☐ D AWS Trusted Advisor

◆ PAUSE AND DECIDE

Where should they look?

- A** Amazon CloudWatch dashboards
Shows their own resources' metrics, which they have already checked and found normal.
- C** AWS CloudTrail event history
Shows API calls made in the account. Nobody in the company did anything; this is not a change they made.

- B** The AWS Health Dashboard
Reports AWS service events and, in the account view, those affecting the company's specific resources and maintenance.
- D** AWS Trusted Advisor
Best-practice recommendations. It does not report AWS service incidents.

THE SIGNAL

Their own metrics are normal, which rules out every service that watches their resources. The problem is on AWS's side.

A finance team asks for two things. First, a list of any EC2 instances running at consistently low utilisation, with a recommendation for a smaller instance type. Second, a broader review flagging idle load balancers, unassociated Elastic IP addresses and security groups with unrestricted ports.

They want the right service for each.

Which two services should the team use? (Choose two.)

- ☐ A AWS Compute Optimizer
- ☐ B AWS Trusted Advisor
- ☐ C AWS Config
- ☐ D Amazon CloudWatch
- ☐ E AWS Cost Explorer

◆ PAUSE AND DECIDE

Which two services should the team use? (Choose two.)

- A** AWS Compute Optimizer
Analyses utilisation and recommends a specific smaller instance type. That is precisely the first request.
- B** AWS Trusted Advisor
Idle load balancers, unassociated Elastic IPs and open security groups are all named Trusted Advisor checks.
- C** AWS Config
Evaluates resources against rules you define. It makes no cost or best-practice recommendations of its own.
- D** Amazon CloudWatch
Supplies the underlying utilisation metrics, but it recommends nothing.
- E** AWS Cost Explorer
Shows what you spent and where. It does not identify idle resources or open ports.

THE SIGNAL

Recommend a specific size is Compute Optimizer. A broad sweep across resource types is Trusted Advisor.

What to carry out of this module

- ◆ **How is it behaving is CloudWatch. Who did it is CloudTrail. What was it set to is Config.**
- ◆ A named user or the word who always means CloudTrail. A threshold on a number always means CloudWatch.
- ◆ **An alarm plus SNS is the standard notification pairing.** One detects, the other delivers.
- ◆ Session Manager gives shell access with no open ports and no keys. It replaces the bastion host.
- ◆ The Health Dashboard reports AWS's health. CloudWatch reports yours.

ON THE EXAM

Trusted Advisor has six categories since 2023. Older material says five, and those five are still correct.